A070 v0.2.2 | Axial torsion propagation gate

18 October 2026 · DEVELOPMENT / SYNTHETIC ONLY · NOT PKLSA CERTIFIED

Scope: geometric correction to the v0.2.1 broad-angle synthetic blind reference. The proposed SST light pulse must travel parallel to the cylindrical vortex filament. The transverse azimuthal oscillation is not a transverse displacement of the vortex axis. The original v0.2.1 frozen artifacts are preserved.

1. Geometry is fixed before any blind classification

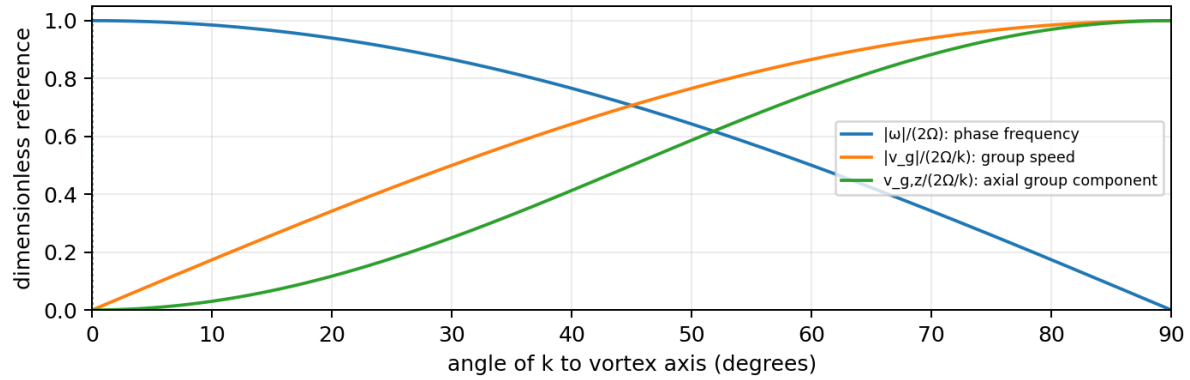
> Filament axis: e_z | wavevector: $k = k_z e_z$ | angle(k, axis): 0° | angle(k, endcap plane): 90°.

This does not mean 90° to the curved cylinder wall. Along that surface the wavevector is tangent. Both +z and −z are admissible propagation directions. The earlier 50° reference has no status as an SST light prediction.

2. Analytic uniform-rotation inertial-wave negative control

> $\omega_I = \pm 2\,\Omega\, k_z/|k|$; $v_{g,I} = \pm 2\,\Omega\,[e_z/|k| - (k_z\, k)/|k|^3]$.

Exactly axial k gives $|\omega_I|=2|\Omega|$, yet $v_{g,I}=0$. A nonzero frequency is therefore not proof of pulse transport along the filament. At a transverse wavevector, the frequency vanishes but the limiting group velocity can be axial. This distinction is the key correction.



The full angle scan remains useful as a diagnostic of the known inertial wave, but not as the physical acceptance criterion of the axial SST photon hypothesis.

3. Quiescent and advected Euler references

> $u_0=0$: $\partial_t\, \delta u_\varphi = 0$. $u_0=U_z e_z$: $(\partial_t + U_z\, \partial_z)\, \delta u_\varphi=0$.

These exact linearized controls apply to a pure divergence-free axisymmetric azimuthal disturbance in an otherwise quiescent, or uniformly axially advecting, incompressible inviscid flow. Advected phase motion is not an intrinsic wave. A swirling finite-core background is more complex: pressure, radial/axial flow and filament deformation must be resolved by the full Euler/PKLSA equations.

Blinding, diagnostics, and physical limits

4. Two separately frozen blind suites

Suite	Intended purpose	Status
v0.2.1 | 5 masked cases	7 angle × 5 k bulk inertial-mode classifier	Old lock preserved
v0.2.2 | 4 masked cases	Axial phase separation at fixed k || filament	New independent lock
Native finite-core PKLSA	Nonzero intrinsic axial packet energy transport	NOT RUN

The 4 synthetic axial masks represent a stationary case, passive advective phase, a manually imposed traveling phase, and a nonmatching pattern. The labels are absent from the BLIND archive and maintained only in a separate REVEAL archive. Successful identification of a prescribed traveling phase is NOT evidence of an emergent torsional Euler eigenmode.

5. Observable required of a real axial pulse

$v_{intrinsic,z} = v_{packet,z} - U_z$ (evaluated in the local filament-comoving frame).

The necessary but insufficient gate is a statistically resolved and spatially/temporally converged nonzero intrinsic packet-energy transport along the core. A phase-frequency scan alone is insufficient. Independently classify Kelvin bending, passive advection, bulk inertial modes, shear-induced perturbations, and pressure-coupled torsion.

6. Results and gates

25/25 regressions PASS; original 0.2.1 frozen SHA retained; the new axial classification is independently hashed and reproducible. These are mathematical and synthetic workflow results. Physical photon claims: UNRESOLVED.

AX0 geometry: PASS (analytic). AX1 quiescent/advected nulls: PASS (analytic). AX2 inertial group degeneracy: PASS (analytic). AX3 masked synthetic axial classifier: PASS. AX4 native finite-core axial torsion: NOT RUN. AX5 native energy transport: NOT RUN. AX6 Swirl-Clock/rod closure: NOT RUN. AX7 electromagnetic observations: NOT RUN.

7. Time and distance definitions

SST uses universal absolute time t for background Euler dynamics. Atomic local time τ_i is defined by independent Swirl-Clock phase; operational distance ℓ_i is a separate atomic/radar-measurement outcome. Neither τ_i nor ℓ_i, nor the target speed c, is used to tune the blind axial classifier.

References

H. P. Greenspan (1968), The Theory of Rotating Fluids, Cambridge University Press. ISBN 9780521043885. G. K. Batchelor (1967), An Introduction to Fluid Dynamics, Cambridge University Press. DOI: 10.1017/CBO9780511800955.